24EECE2001 – Electronic Devices and Circuits

AMPLIFIER FREQUENCY RESPONSE

Prepared by:
Prof. Madhavi Tatineni
Department of EECE
GITAM School of Core Engineering
Hyderabad

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Adel S. Sedra, Kenneth C. Smith, Arun N. Chandorkar, Microelectronic Circuits, 7/e, Oxford University Press, 2017

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Syllabus

MODULE 1 SEMICONDUCTORS Intrinsic Semiconductors, Doped Semiconductors, Current Flow in Semiconductors, The pn Junction with Open-Circuit Terminals (Equilibrium), The pn Junction with Applied Voltage

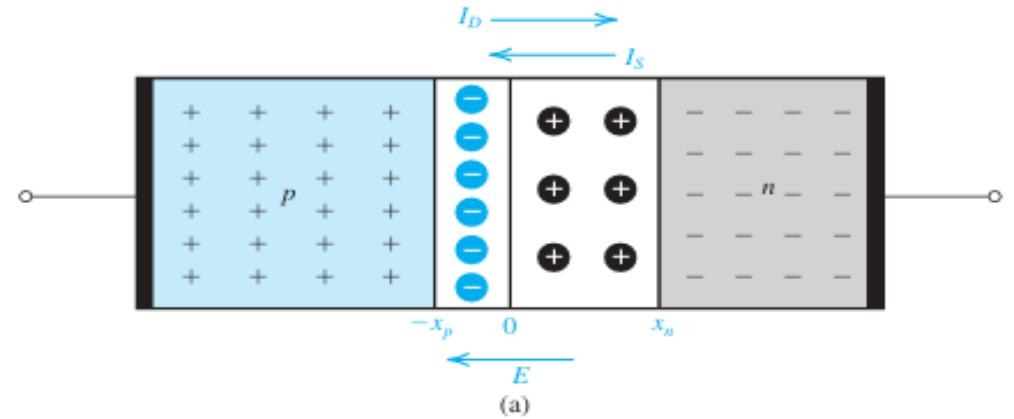
MODULE 2 DIODES AND DIODE CIRCUITS The Ideal Diode, Terminal Characteristics of Junction Diodes, Modeling the Diode Forward Characteristic, Operation in the Reverse Breakdown Region—Zener Diodes, Rectifier Circuits, Limiting and Clamping Circuits

MODULE 3 BIPOLAR JUNCTION TRANSISTORS Device Structure and Physical Operation, Current–Voltage Characteristics, BJT Circuits at DC, Applying the BJT in Amplifier Design, Small-Signal Operation and Models, Basic BJT Amplifier Configurations. Transistor Breakdown and temperature effects

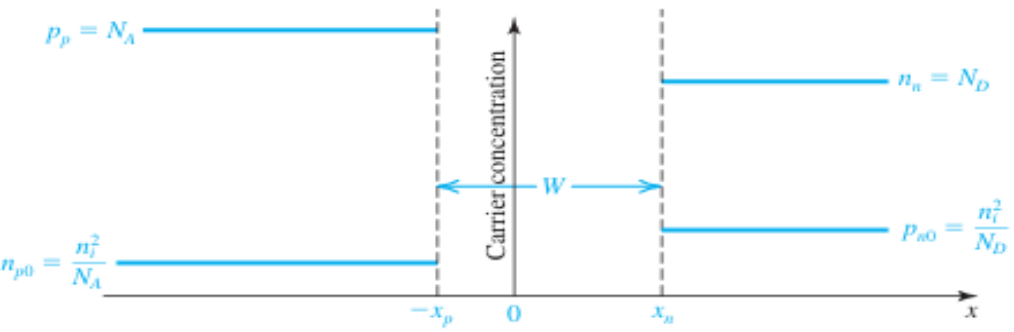
MODULE 4 MOS FIELD EFFECT TRANSISTORS Device Structure and Physical Operation, Current—Voltage Characteristics, MOSFET Circuits at DC, Applying the MOSFET in Amplifier Design, Small-Signal Operation and Models, Basic MOSFET Amplifier Configurations, Biasing in MOS Amplifier Circuits, Discrete-Circuit MOS Amplifiers

MODULE 5 AMPLIFIER FREQUENCY RESPONSE Review of Transfer function, Frequency Response, Pole-Zero Concepts, Transfer function and AC Analysis of a First Order RC Circuit, Capacitive Effects in a PN Junction, Internal Capacitive Effects and the High-Frequency Model of the MOSFET, High-Frequency Response of the Common Source Amplifiers, High Frequency Response of Common Gate Amplifiers, High Frequency Response of Source Followers

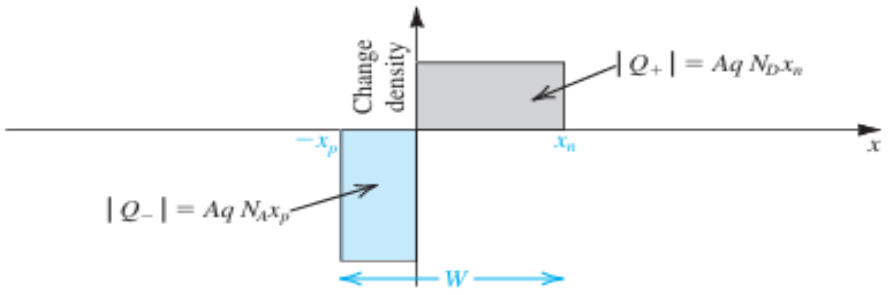
Width of the charge stored in the depletion region



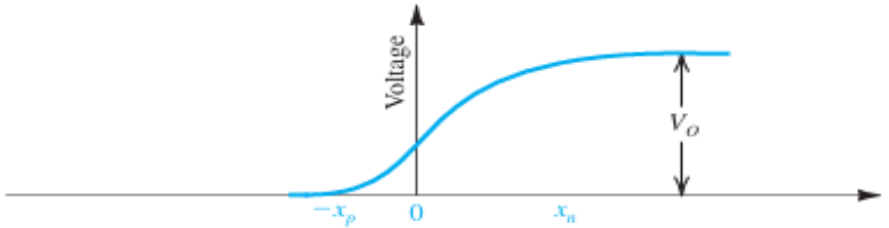
(a)



(b)



(c)



(d)

minority carrier concentrations in both sides are denoted by n_{p0} and p_{n0}

To uncover the same amount of charge, the depletion layer will extend deeper into the more lightly doped material

Depletion region extends in both the *p* and *n* materials and that equal amounts of charge exist on both sides (and Q+ and Q-)

If we denote the width of the depletion region in the *p* side by x_p and in the *n* side by x_n, the magnitude of the charge on the *n* side of the junction can be expressed as

$$|Q_+| = qAx_nN_D \quad (1) \qquad |Q_-| = qAx_pN_A \quad (2)$$

(a) A *pn* junction with the terminals open circuited. **(b)** Carrier concentrations; note that $N_A \gg N_D$. **(c)** The charge stored in both sides of the depletion region; $Q_J = |Q_+| = |Q_-|$. **(d)** The built-in voltage V_0.

charge equality condition: $qAx_nN_D = qAx_pN_A$

$$\frac{x_n}{x_p} = \frac{N_A}{N_D} \qquad (3)$$

Usually, one side of the junction is doped much more heavily than the other, with the result that the depletion region exists almost entirely on one side (the lightly doped side).

The width W of the depletion layer can be shown to be given by

$$W = x_n + x_p = \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)V_0} \qquad (4)$$

Where ε_s is the electrical permittivity of silicon = 11.7 ε_0 = 11.7 x 8.85 x10^{-14}F/cm = 1.04x10^{-12} F/cm.

Typically W is in the range 0.1 μm to 1 μm.

To express in terms of W, this equation can be rewritten as

$$x_n N_D - x_p N_A + N_A x_n - N_A x_n = 0;\ x_n (N_A + N_D) = N_A (x_n + x_p) \qquad (5)$$

$$x_n = \frac{N_A}{N_{A+N_D}} W \qquad (6)$$

$$x_p = \frac{N_D}{N_{A+N_D}} W \qquad (7)$$

The charge stored on either side of the depletion region can be expressed in terms of *W* by

$$Q_J = |Q_+| = |Q_-|$$

$$Q_J = Aq\left(\frac{N_A N_D}{N_A + N_D}\right) W \qquad (8)$$

By substituting for *W* from the equation (4), Q_J can be written as

$$Q_J = A\sqrt{2\varepsilon_s q\left(\frac{N_A N_D}{N_A + N_D}\right) V_0} \qquad (9)$$

Capacitive Effects in the pn Junction

Depletion or Junction Capacitance – Exists when the PN Junction is reverse biased
Diffusion Capacitance – Exists when the PN Junction is forward biased

Depletion or Junction Capacitance:

When a pn junction is reverse biased with a voltage V_R, the charge stored on either side of the depletion region is given by

$$Q_J = A\sqrt{2\varepsilon_s q\left(\frac{N_A N_D}{N_A+N_D}\right)(V_0+V_R)} = \alpha\sqrt{V_0+V_R}$$

where α is given by $\alpha = A\sqrt{2\varepsilon_s q\left(\frac{N_A N_D}{N_A+N_D}\right)}$

Depletion or Junction capacitance $\boldsymbol{C_j}$ that relates the change in the charge $\boldsymbol{Q_J}$ to a change in the voltage V_R,

$$C_j = \left.\frac{dQ_J}{dV_R}\right|_{V_R=V_Q} \qquad C_j = \frac{\alpha}{2\sqrt{V_0+V_R}}$$

The value of at zero reverse-bias can be obtained as $C_{j0} = \frac{\alpha}{2\sqrt{V_0}}$

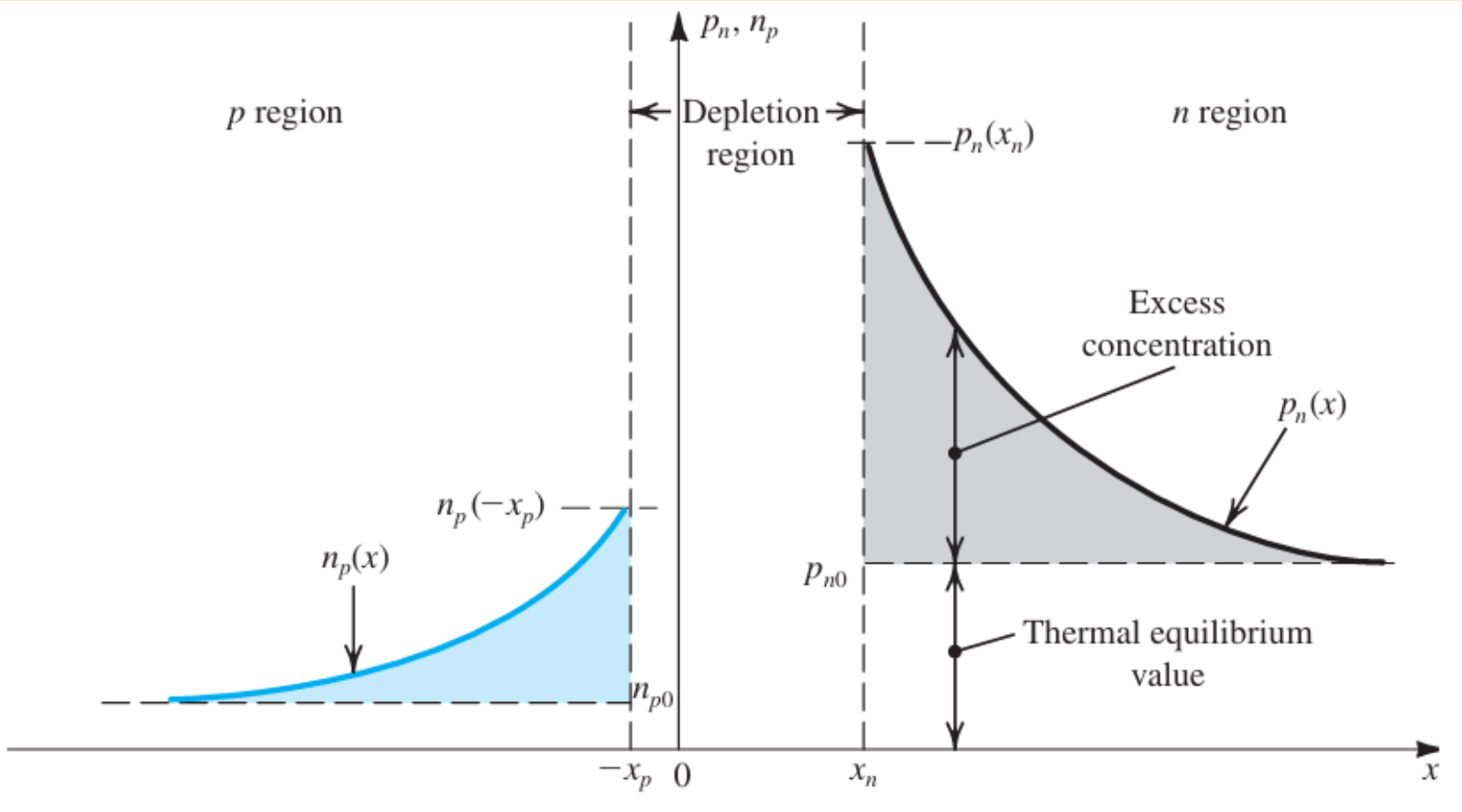
$$C_j = \frac{\alpha}{2\sqrt{\left[(1+\frac{V_R}{V_0})\right]V_0}} = \frac{\alpha}{2\sqrt{V_0}\sqrt{\left[(1+\frac{V_R}{V_0})\right]}} = \frac{C_{j0}}{\sqrt{\left[(1+\frac{V_R}{V_0})\right]}}$$

There is another type of pn junction in which the carrier concentration is made to change gradually from one side of the junction to the other. To allow for such a graded junction, the formula for the junction capacitance can be written in the more general form

$$C_j = \frac{C_{j0}}{\left(1 + \frac{V_R}{V_0}\right)^m}$$

where m is a constant called the **grading coefficient**, whose value ranges from 1/3 to 1/2 depending on the manner in which the concentration changes from the p to the n side.

Diffusion Capacitance



Minority-carrier distribution in a forward-biased pn junction

We know that steady-state concentration at the edge of the depletion region will be $P_n(x_n) = P_{n0}e^{V/V_T}$ (1)

and hole–diffusion current density at $x = x_n$, $J_p(x_n) = q\left(\frac{D_P}{L_p}\right)P_{n0}\left(e^{V/V_T} - 1\right)$ (2)

The excess hole charge stored in the *n* region can be found from the shaded area under the exponential as

$Q_P = charge\ q * Cross\ sectional\ area\ A *$ shaded area under the $P_n(x) curve$

$$Q_P = Aq[P_n(x_n) - P_{n0}]L_P \qquad (3)$$

$$= AqL_P P_{n0}\left[e^{V/V_T} - 1\right] = Aq\frac{D_P}{L_P}\frac{L_P}{D_P}L_P P_{n0}\left[e^{V/V_T} - 1\right] \qquad (4)$$

We know that

$$I_P = Aq\frac{D_P}{L_P}P_{n0}\left[e^{V/V_T} - 1\right] \qquad (5)$$

By considering equation (5), eqn (4) can be written as $Q_P = I_P\frac{L_P^2}{D_P} = I_P\tau_P$

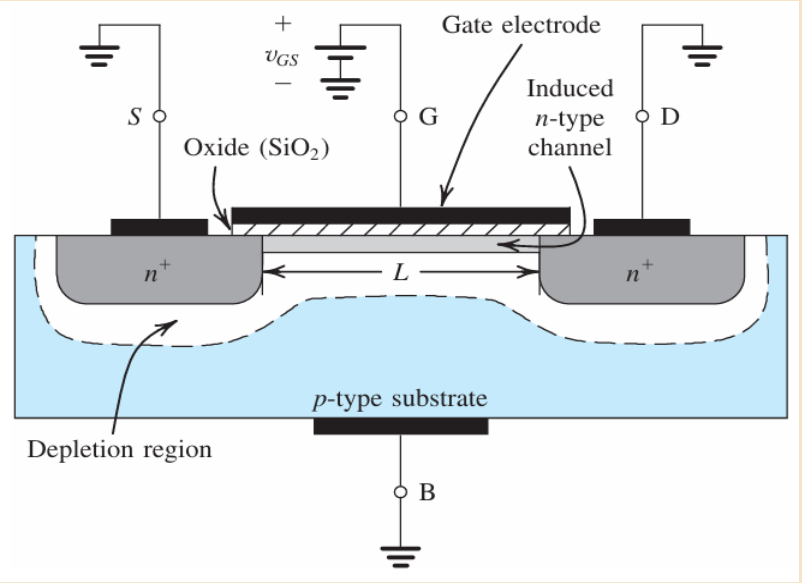
The time constant τ_P is known as the excess minority carrier (hole) lifetime. It is the average time it takes for a hole injected into the n region to recombine with a majority electron.

Similarly, we can write the equation for Q_n as $Q_n = I_n\tau_n$

Total charge, $Q = I_P\tau_P + I_n\tau_n = (\tau_T I)$ Diffusion Capacitance, $C_d = \frac{dQ}{dV} = \frac{\tau_T}{V_T}I$

where $\boldsymbol{I}$ is the forward-bias current. Note that $\boldsymbol{C_d}$ is directly proportional to the forward current $\boldsymbol{I}$ and thus is negligibly small when the diode is reverse biased. Also note that to keep $\boldsymbol{C_d}$ small, the transit time $\boldsymbol{\tau_T}$ must be made small, an important requirement for a pn junction intended for high-speed or high-frequency operation.

Internal Capacitive Effects and the High-Frequency Model of the MOSFET



There are basically two types of internal capacitance in the MOSFET:

The gate capacitive effect: The gate electrode (polysilicon) forms a parallel-plate capacitor with the channel, with the oxide layer serving as the capacitor dielectric.

The source-body and drain-body depletion-layer capacitances: These are the capacitances of the reverse-biased pn junctions formed by the n+ source region and the p-type substrate and by the n+ drain region and the substrate.

Also Capacitances exist between the four terminals of the MOSFET, G, D, S and B.

There will be five capacitances in total: C_{gs}, C_{gd}, C_{gb}, C_{sb}, and C_{db}, where the subscripts indicate the location of the capacitances in the model.

The Gate Capacitive Effect:

The gate capacitive effect can be modeled by the three capacitances C_{gs}, C_{gd}, C_{gb}.

In Triode region:

When the MOSFET is operating in the triode region at small v_{DS}, the channel will be of uniform depth. The gate-channel capacitance will be WLC_{ox} and can be modeled by dividing it equally between the source and drain ends; thus,

$$C_{gs} = C_{gd} = \frac{1}{2} WL\, C_{ox} \quad \text{(triode region)}$$

In Saturation region:

When the MOSFET operates in saturation, the channel has a tapered shape and is pinched off at or near the drain end.

$$\left.\begin{aligned} C_{gs} &= \frac{2}{3} WL\, C_{ox} \\ C_{gd} &= 0 \end{aligned}\right\} \quad \text{(saturation region)}$$

In Cutoff region:

When the MOSFET is cut off, the channel disappears, and thus $C_{gs} = C_{gd} = 0$. capacitance WLC_{ox} exists between gate and body.

$$\left.\begin{aligned} C_{gs} &= C_{gd} = 0 \\ C_{gb} &= WL\, C_{ox} \end{aligned}\right\} \quad \text{(cutoff)}$$

The Junction Capacitances:

The depletion-layer capacitances of the two reverse-biased pn junctions formed between each of the source and the drain diffusions and the body.

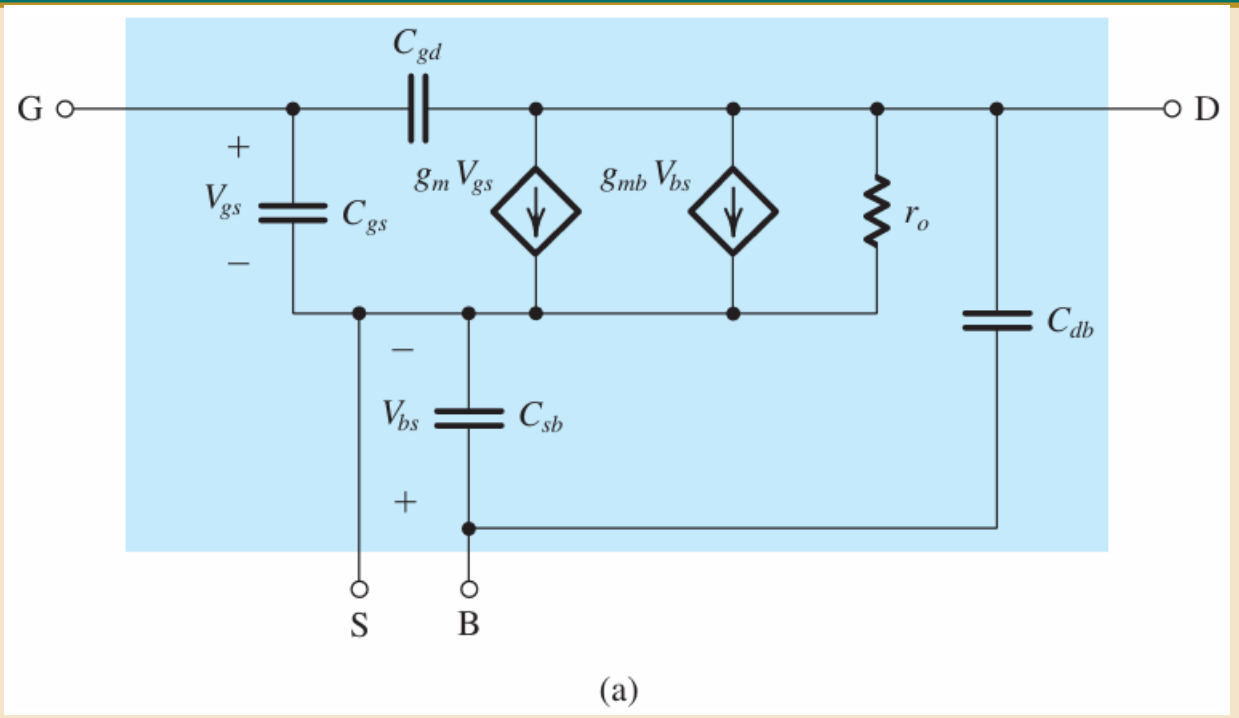
We know that $$C_j = \frac{C_{j0}}{\sqrt{1 + \frac{V_R}{V_0}}}$$

for the source diffusion, we have the source-body capacitance, C_{sb}, $$C_{sb} = \frac{C_{sb0}}{\sqrt{1 + \frac{V_{SB}}{V_0}}}$$

where C_{sb0} is the value of C_{sb} at zero body-source bias, V_{SB} is the magnitude of the reverse bias voltage, and V_0 is the junction built-in voltage (0.6 V to 0.8 V). Similarly, for the drain diffusion, we have the drain-body capacitance C_{db}, $$C_{db} = \frac{C_{db0}}{\sqrt{1 + \frac{V_{DB}}{V_0}}}$$

where C_{db0} is the capacitance value at zero reverse-bias voltage, and V_{DB} is the magnitude of this reverse-bias voltage.

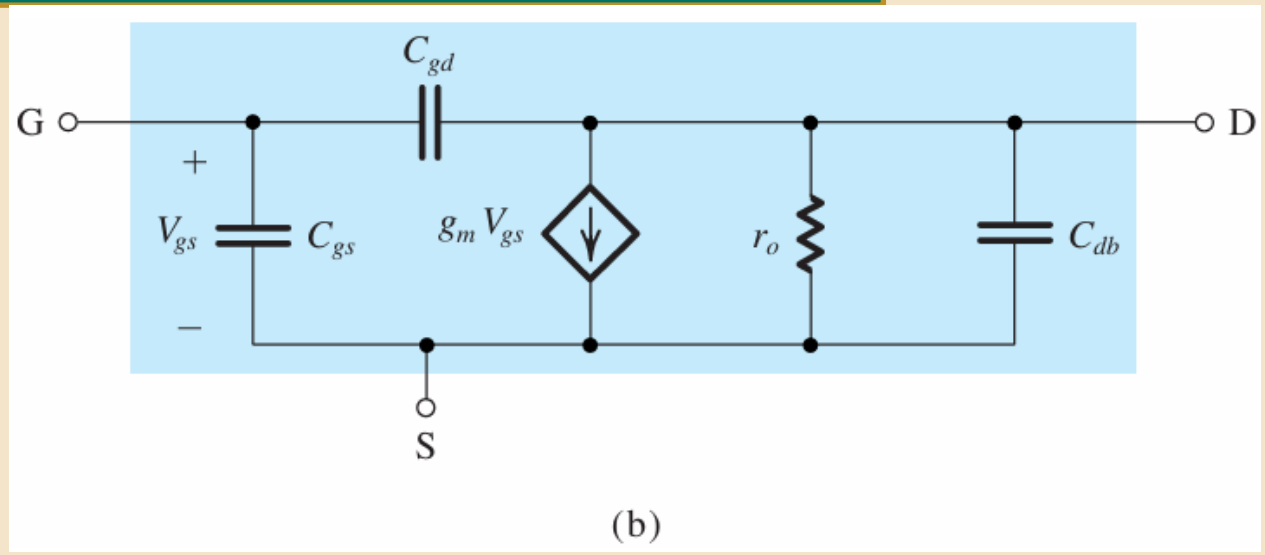
The High-Frequency MOSFET Model



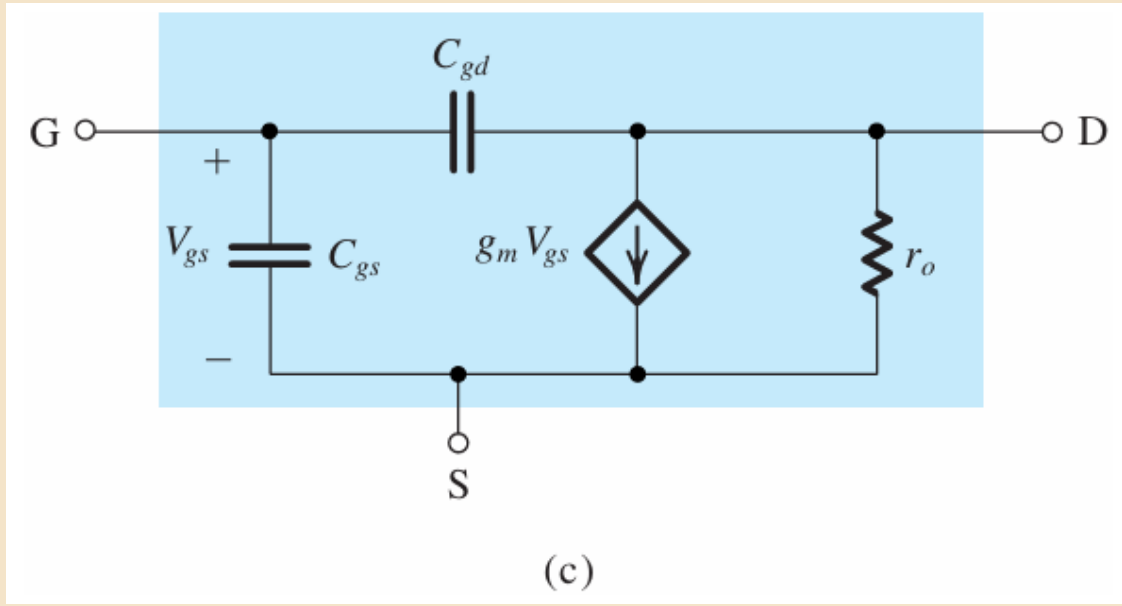
High-frequency, equivalent-circuit model for the MOSFET

C_{gs}: Gate-to-source capacitance.
C_{gd}: Gate-to-drain capacitance.
C_{sb}: Source-to-body junction capacitance.
C_{db}: Drain-to-body junction capacitance.
$g_m V_{gs}$: Drain current controlled by gate-to-source voltage.
$g_{mb} V_{bs}$: Body-effect-dependent current source.
r_o: Small-signal output resistance due to channel-length modulation.

C_{gd}, although small, plays a significant role in determining the high-frequency response of amplifiers and thus must be kept in the model.



The equivalent circuit for the case in which the source is connected to the substrate (body).



The equivalent-circuit model of (b) with C_{db} neglected

The MOSFET Unity-Gain Frequency (f_T)

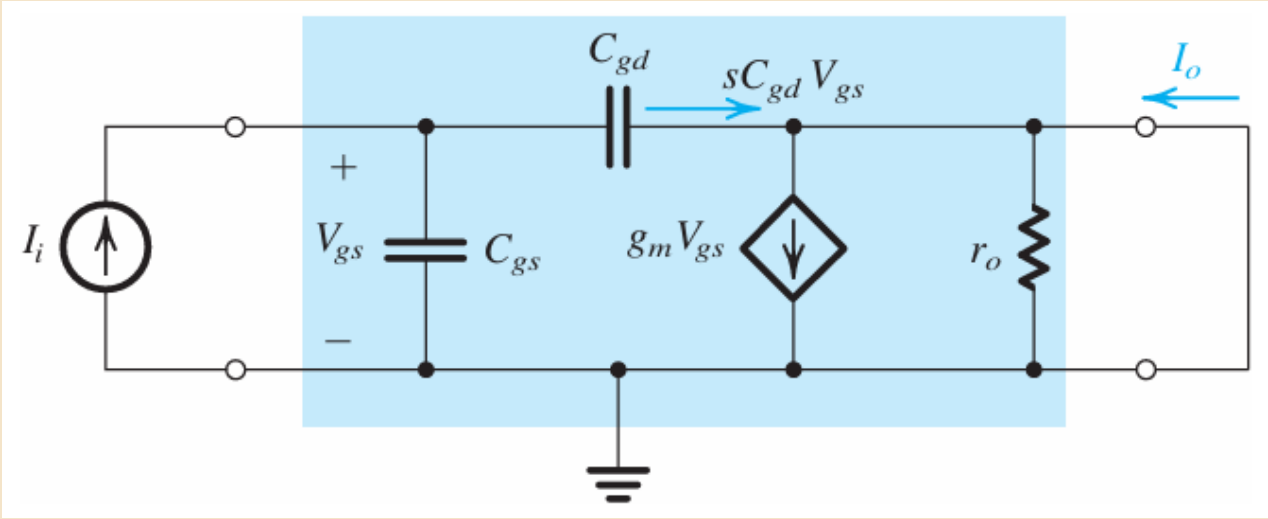
A figure of merit for the high-frequency operation of the MOSFET as an amplifier is the unity-gain frequency, $\boldsymbol{f_T}$, also known as the transition frequency, which gives rise to the subscript T. **This is defined as the frequency at which the short-circuit current-gain of the common-source configuration becomes unity.**

At low frequencies, the MOSFET can provide high current gain. As the frequency increases, the current gain decreases. At the unity-gain frequency, $\boldsymbol{f_T}$, the current gain becomes 1.

Importance of $\boldsymbol{f_T}$:

A larger $\boldsymbol{f_T}$ (unity-gain frequency) is generally an advantage for a MOSFET used in high-frequency applications.
Larger $\boldsymbol{f_T}$:The MOSFET can provide current gain at higher frequencies, making it suitable for RF and high-speed circuits.
Smaller $\boldsymbol{f_T}$: The current gain falls to unity at a lower frequency, limiting its high-frequency performance.

Determining the short-circuit current gain I_o/I_i



To determine the short-circuit current gain, the input is fed with a current-source signal $\boldsymbol{I_i}$ and the output terminals are short-circuited.

Current flowing through the capacitor, Cgd = $\frac{V_{gs}}{1/sC_{gd}} = sC_{gd}V_{gs}$

Current in the short circuit, $\boldsymbol{I_0}$ is given as $I_0 = g_mV_{gs} - sC_{gd}V_{gs}$

As C_{gd} is very small, we can neglect the second term and $I_0 \approx g_mV_{gs}$

From Fig, $I_i = sC_{gs}V_{gs} + sC_{gd}V_{gs} = s(C_{gs} + C_{gd})V_{gs}$

Therefore the short-circuit current gain, $\frac{I_0}{I_i} = \frac{g_m}{s(C_{gs}+C_{gd})}$

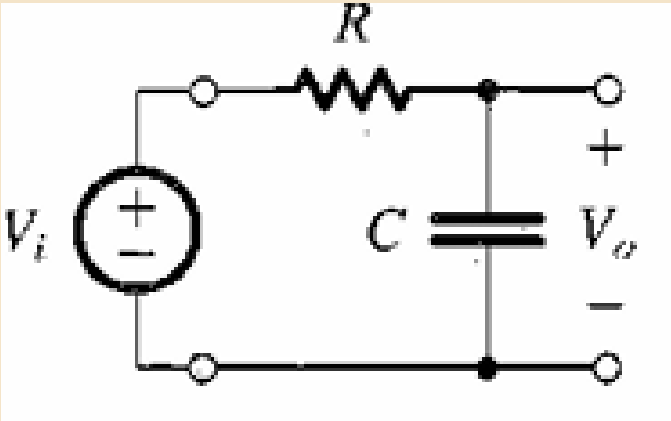
By substituting s = jω, the magnitude of the current gain becomes unity at ω = ω_T.

$$\left|\frac{I_0}{I_i}\right| = 1 \; at \; \omega = \omega_T \qquad \frac{g_m}{\omega_T(C_{gs}+C_{gd})} = 1$$

Unity-gain frequency, $f_T = \omega_T/2\pi$ is, $f_T = \frac{g_m}{2\pi(C_{gs}+C_{gd})}$

Since f_T is proportional to g_m and inversely proportional to the MOSFET internal capacitances, the higher the value of f_T, the more effective the MOSFET becomes as an amplifier.

RC Low Pass Filter



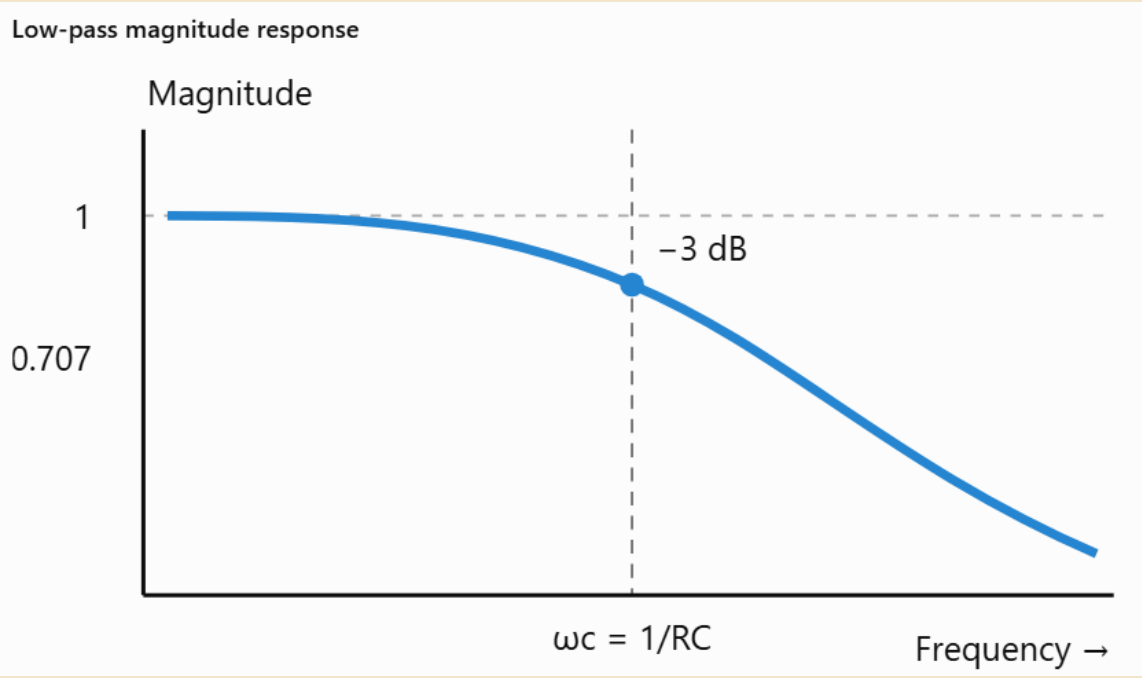
Low-pass filter (LPF): Allows low frequencies to pass and attenuates high frequencies.

The **transfer function** is the ratio of output voltage to input voltage.
A **transfer function** tells us how a system changes an input signal to produce an output signal.

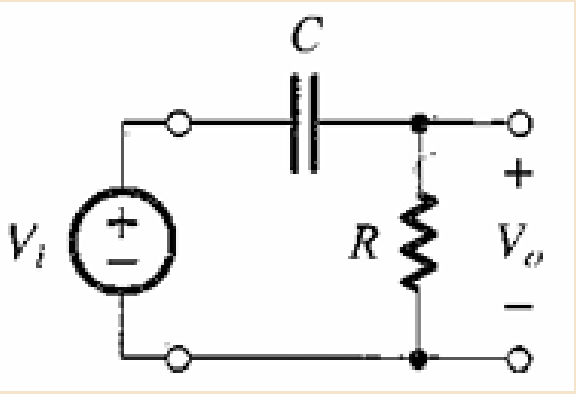
$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{1}{1 + sRC}$$

For sinusoidal signals, put s=jω): $H(j\omega) = \frac{1}{1 + j\omega RC}$

Magnitude Response: $|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}}$



RC High Pass Filter



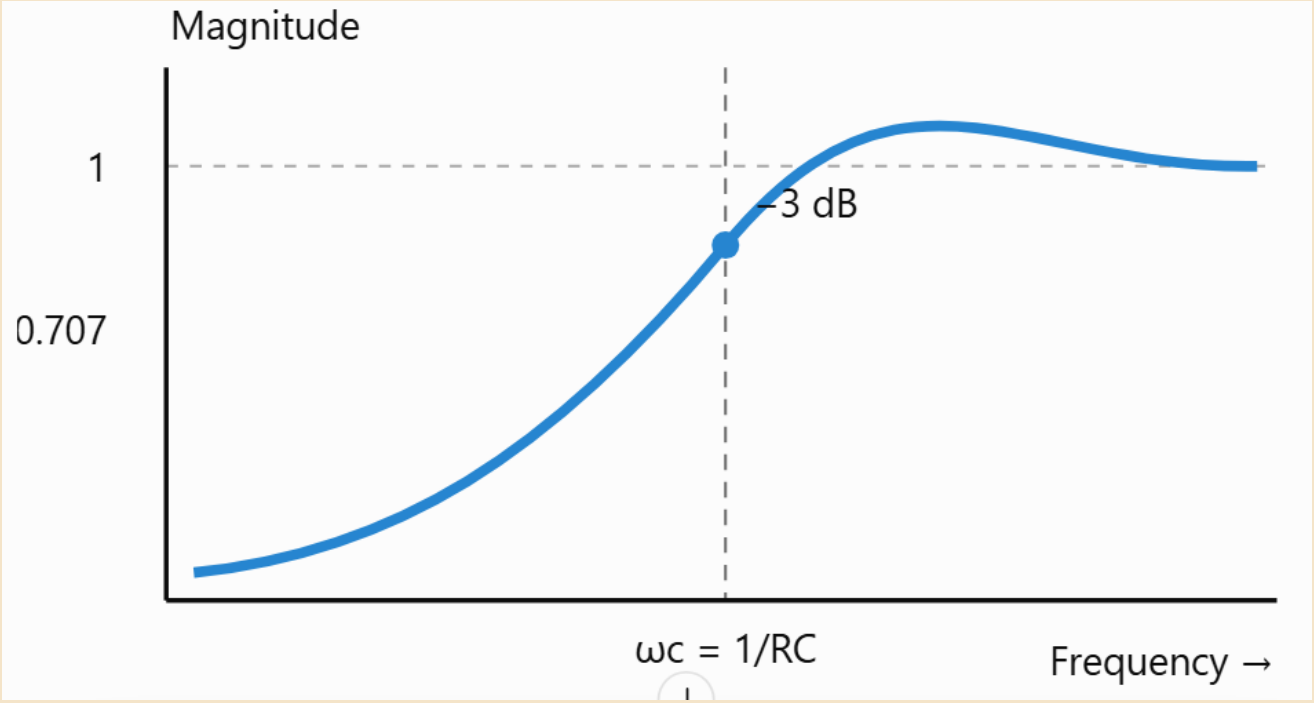
$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{sRC}{1+sRC}$$

$$H(j\omega) = \frac{j\omega RC}{1+j\omega RC}$$

Magnitude response $$|H(j\omega)| = \frac{\omega RC}{\sqrt{1+(\omega RC)^2}}$$

For both filters, the cut-off frequency is $$f_c = \frac{1}{2\pi RC}$$

At ($f=f_c$), the output magnitude is 0.707 times the input magnitude, which corresponds to -3dB.



magnitude response using poles and zeros

$$H(s) = \frac{\text{Numerator}}{\text{Denominator}}$$

Poles are the values of s that make a system's transfer-function denominator zero, while zeros are the values of s that make its numerator zero; together they determine stability, frequency response, and transient behavior of linear systems.

A pole frequency corresponds to a corner frequency at which the slope of the magnitude curve decreases by 20 dB/decade, and a zero corresponds to a corner frequency at which the slope increases by 20 dB/decade.

RC Low-Pass Filter $H(s) = \frac{1}{1 + sRC}$

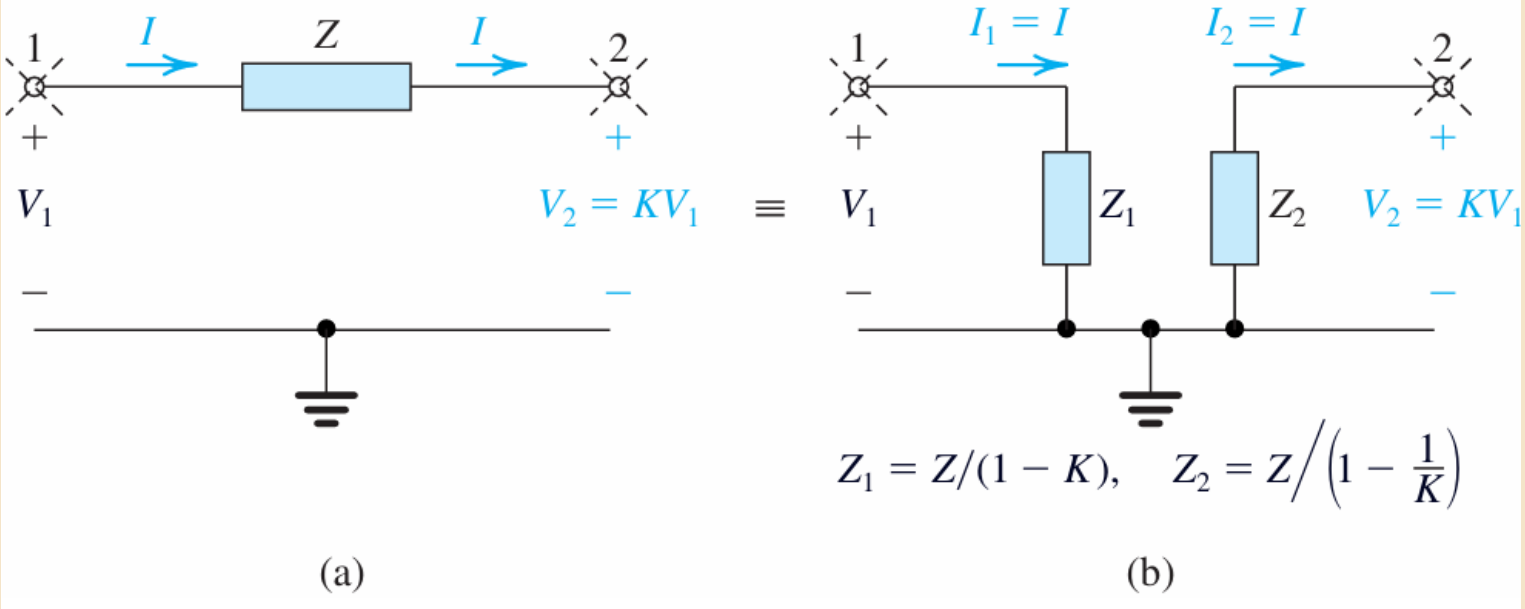
The numerator is 1, which never becomes zero. Therefore, the filter has no finite zero.
The denominator has a pole at s = -1/RC.

RC High-Pass Filter $H(s) = \frac{sRC}{1 + sRC}$

Making Numerator zero, s = 0. This is the zero of the high pass filter.
Making denominator equal to zero, it has a pole at s= -1/RC

Miller's Theorem

Replacing the bridging capacitance by an equivalent input capacitance based on a general theorem known as Miller's theorem,
Miller's theorem states that impedance Z can be replaced by two impedances: Z_1 between node 1 and ground and Z_2 connected between node 2 and ground.



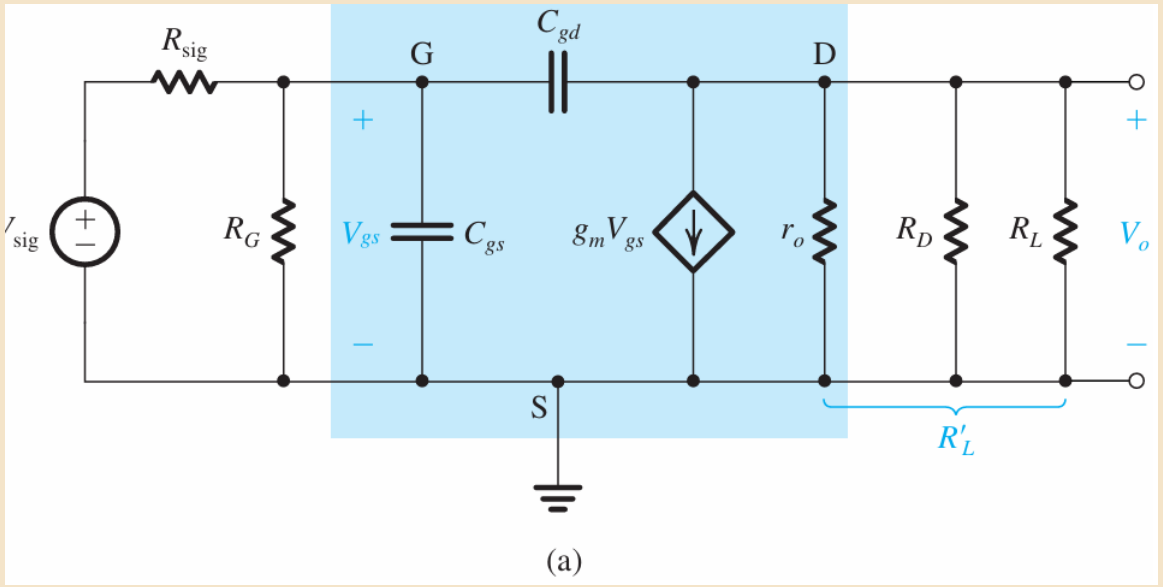
$$I_1 = \frac{V_1}{Z_1} = I = \frac{V_1 - KV_1}{Z}$$

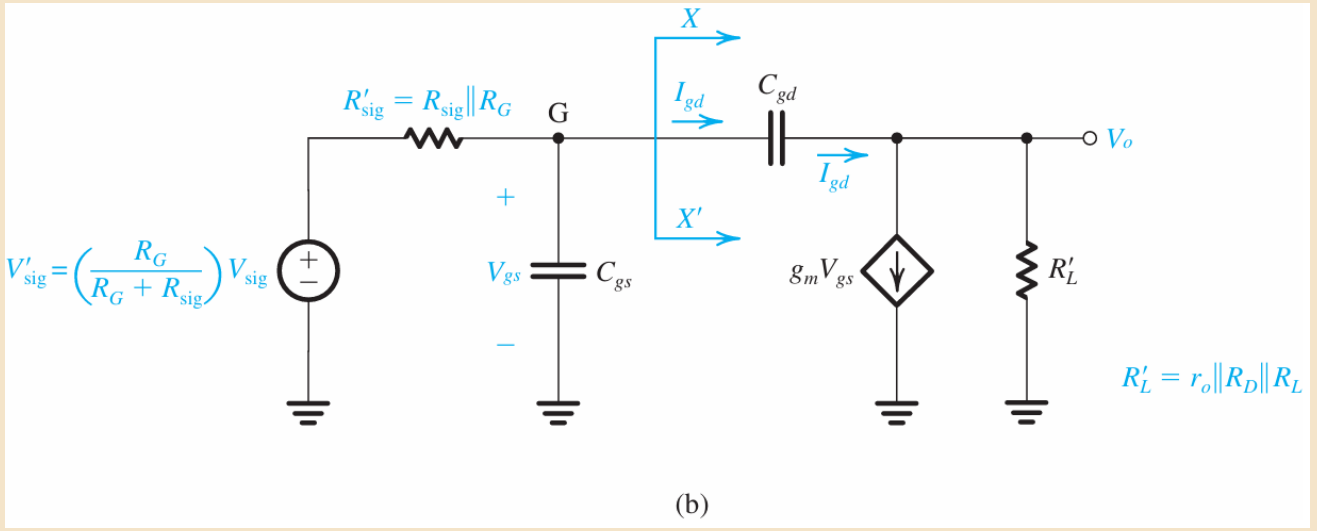
$$Z_1 = Z/(1-K)$$

$$I_2 = \frac{0 - V_2}{Z_2} = \frac{0 - KV_1}{Z_2} = I = \frac{V_1 - KV_1}{Z}$$

$$Z_2 = Z\Big/\left(1 - \frac{1}{K}\right)$$

High-Frequency Response of the CS Amplifier





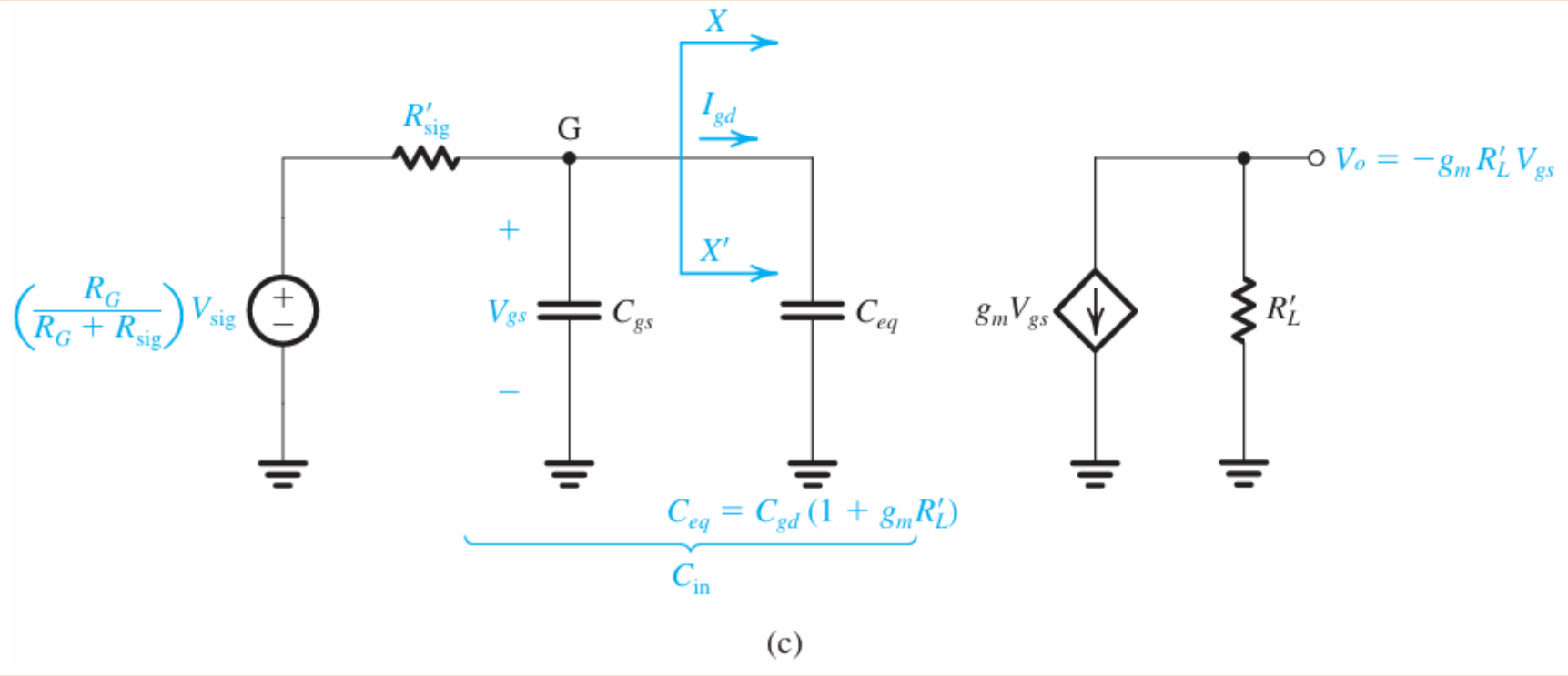
Determining the high-frequency response of the CS amplifier:
(a) equivalent circuit; (b) the circuit of (a) simplified at the input and the output

The midband gain A_M circuit by setting C_{gs} and C_{gd} to zero.

$$A_M = \frac{V_o}{V_{sig}} = -\frac{R_G}{R_G + R_{sig}}(g_m R'_L)$$

$$V_o \simeq -(g_m V_{gs})R'_L = -g_m R'_L V_{gs}$$

$$R'_L = r_o \| R_D \| R_L$$



$$C_{eq} = C_{gd}(1 + g_m R'_L)$$

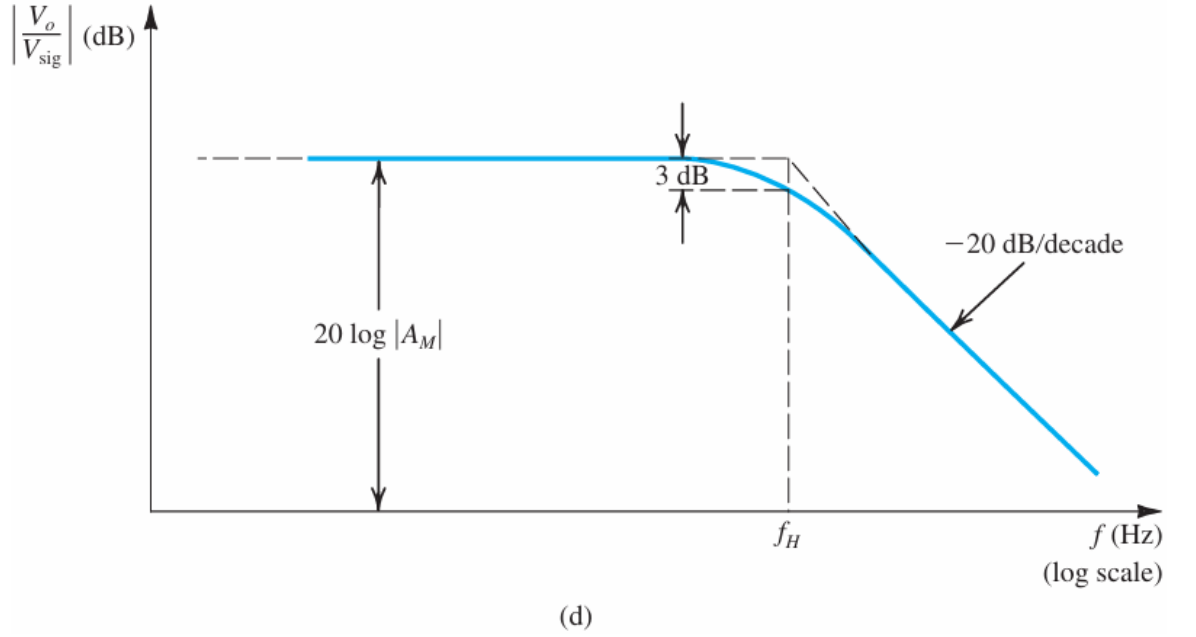
$$f_H = \frac{1}{2\pi C_{in} R'_{sig}}$$

$$C_{in} = C_{gs} + C_{eq} = C_{gs} + C_{gd}(1 + g_m R'_L)$$

$$R'_{sig} = R_{sig} \parallel R_G$$

(c) the equivalent circuit with C_{gd} replaced at the input side with the equivalent capacitance C_{eq}

C_{gd} gives rise to a much larger capacitance C_{eq}, which appears at the amplifier input. The multiplication effect that C_{gd} undergoes comes about because it is connected between circuit nodes **g** and **d**, whose voltages are related by a large negative gain ($-g_m R_L'$). This effect is known as the **Miller effect**, and ($1 + g_m R_L'$) is known as the **Miller multiplier**.



(d)

(d) the frequency response plot, which is that of a low-pass, single-time-constant circuit

The high-frequency response will be that of a low-pass STC network with a 3-dB frequency f_H determined by the time constant $C_{in}R'_{sig}$

1. The upper 3-dB frequency is determined by the interaction of $R'_{sig} = R_{sig} \| R_G$ and $C_{in} = C_{gs} + C_{gd}(1 + g_m R'_L)$. Since the bias resistance R_G is usually very large, it can be neglected, resulting in $R'_{sig} \simeq R_{sig}$, the resistance of the signal source. It follows that a large value of R_{sig} will cause f_H to be lowered.
2. The total input capacitance C_{in} is usually dominated by C_{eq}, which in turn is made large by the multiplication effect that C_{gd} undergoes. Thus, although C_{gd} is usually a very small capacitance, its effect on the amplifier frequency response can be very significant as a result of its multiplication by the factor $(1 + g_m R'_L)$, which is approximately equal to the midband gain of the amplifier. This is the Miller effect, which causes the CS amplifier to have a large total input capacitance C_{in} and hence a low f_H.

3. To extend the high-frequency response of a MOSFET amplifier, we have to find configurations in which the Miller effect is absent or at least reduced.

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